

Product Requirements Document (PRD): ECO-FLOAT

Team: PUREPATH

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1. Executive Summary

The ECO-FLOAT is an autonomous, floating robotic platform designed to address a critical operational blind spot in aquatic monitoring: the "middle ground" between expensive, low-resolution satellite imagery and labor-intensive manual water sampling. While satellites provide a macro view and manual sampling provides high accuracy, neither offers the continuous, granular data needed to manage rapidly changing small water bodies.

Our goal is to deploy a low-cost (< ₹15,000) unit capable of hyperspectral radiometry to detect chlorophyll levels (indicative of algal blooms) and floating plastic debris in real-time. By combining autonomous robotics, spectral sensing, and Edge AI, ECO-FLOAT aims to democratize water quality data. This allows government bodies, NGOs, and researchers to move from reactive damage control to proactive ecosystem management, visualizing pollution dynamics as they happen rather than waiting for lab reports.

2. Problem Statement

Current methods for monitoring water bodies are inefficient and often reactive rather than proactive. The existing ecosystem forces a compromise between coverage, cost, and resolution:

- **Satellites (Remote Sensing):** While useful for oceanography, they typically offer ~1km resolution (e.g., MODIS) or ~10m (Sentinel-2). This is too coarse for small inland water bodies (lakes, ponds) and river edges, where "mixed pixels" (blurring of land and water data) obscure critical details. Furthermore, satellite data is often rendered useless by cloud cover or long revisit times (2-5 days).

- **Manual Sampling:** This process is logistically heavy, requiring manpower, boats, and strict preservation protocols. Consequently, the data is a "snapshot in time"—often days old by the time results are available—making it impossible to track dynamic events like sudden industrial discharge or flash blooms. There is also the risk of human error in sample collection and transport.
- **Existing Buoys:** Commercial sensing buoys (e.g., YSI/Hach) are highly accurate but cost upwards of ₹50 Lakhs. They are also stationary, meaning they can only measure water quality at a single fixed point. A buoy cannot detect a pollution plume moving just 200 meters away, leaving vast areas unmonitored.

The Result: 8 million tons of plastic enter oceans annually, often originating from unmonitored inland waterways. Simultaneously, algal blooms go undetected until they become eutrophic and toxic, contaminating drinking water sources and affecting over 50 million people dependent on these local water bodies.

3. Product Goals & Objectives

- **Hyper-local Resolution:** Achieve sub-100m² spatial resolution (100x better than satellites) to pinpoint specific point-sources of pollution. This allows authorities to identify not just *that* a lake is polluted, but specifically *which* drain pipe or shoreline activity is causing the issue.
- **Dual-Detection Capability:** Simultaneously monitor biological threats (Chlorophyll-a/Eutrophication) and physical pollution (Floating Plastics). This dual approach is vital as plastics often act as vectors for biofilm and algae growth, meaning these two pollution types are often correlated.
- **Cost Efficiency & Scalability:** Keep the Bill of Materials (BOM) strictly between ₹8,000 - ₹15,000. This disruptive price point enables "swarm deployment," where multiple units can be released to map a large lake simultaneously, offering redundancy and speed that a single expensive unit cannot match.
- **Autonomy:** The bot must navigate pre-defined waypoints via GPS without constant human control. This "set-and-forget" operational model means a single technician can deploy and manage multiple bots from the shore, significantly reducing labor costs.

4. Target Audience

- **Primary: Government Bodies (Municipal Corporations & MoEFCC):** These entities are responsible for maintaining water quality in urban lakes and rivers (e.g., MIDC, PCMC). They need actionable, verifiable data to allocate cleaning budgets effectively and audit the performance of cleaning contractors.
- **Secondary: NGOs & Advocacy Groups:** Organizations like WWF, Greenpeace, or local river foundations need evidence-based data to drive public awareness campaigns, file Public Interest Litigations (PILs), or track the efficacy of community cleanup drives.
- **Tertiary: Research Institutes:** Academic bodies (CSIR, IITs) require high-frequency, granular environmental data to validate satellite models, study microplastic transport mechanisms, or research the correlation between urban runoff and algal blooms.

5. Market Analysis

5.1 Market Segmentation & Growth

The following table outlines the key market segments ECO-FLOAT addresses, highlighting the rapid growth in autonomous systems and the stable demand for water quality data.

Market Segment	Global Value (2024)	Projected Value (2030)	CAGR	Key Market Drivers
Water Quality Monitoring	~\$5.5 Billion	~\$7.4 Billion	~5.1%	• Stricter government regulations (CPCB/EPA). • Shift from periodic to continuous monitoring. • Rising industrial effluent discharge.
Autonomous Surface Vehicles (ASV)	~\$1.2 Billion	~\$2.5 Billion	~14%	• Demand for risk-free data collection.

				• High cost of manned vessel operations.
				• Advances in battery & swarm technology.
Indian Domestic Context	Emerging	High Growth	N/A	• Namami Gange Mission: Huge funding for river health. • Smart Cities: Integration of IoT in urban management. • Plastic Ban: Urgent need to track enforcement.

5.2 Competitive Landscape & Positioning

ECO-FLOAT targets a "Blue Ocean" segment by offering mobility and advanced sensing at a consumer-electronics price point. We are not competing with high-precision lab equipment; we are competing with "no data."

Feature	ECO-FLOAT (Proposed)	Commercial Buoys (YSI/Hach)	Satellite Imagery (Sentinel-2)	Manual Sampling
Cost	< ₹15,000 (disruptive)	> ₹50 Lakhs (High Capex)	Free (Low Res) / High (High Res)	High Opex (Labor/Travel)
Data Type	Hyperspectral + Image	Multi-parameter Probe	Multispectral Imagery	Chemical Lab Analysis
Mobility	Autonomous (Mobile)	Stationary (Fixed)	Global Coverage	Mobile (Human dependent)
Resolution	< 1 meter	Single Point	~10m - 1km	Single Point

Real-Time?	Yes (Edge AI)	Yes	No (2-3 day revisit time)	no(Days/Weeks)
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Strategic Advantage: By pricing significantly lower than incumbents, ECO-FLOAT shifts the procurement model from Capital Expenditure (Capex) to a scalable, almost "disposable" asset class. This allows municipalities to deploy "Swarms" rather than single units, ensuring that if one unit is damaged or stolen, the entire monitoring network does not fail.

6. Functional Requirements

6.1 Sensing & Data Acquisition

The core of the product is the "eye" of the robot. It must utilize spectral analysis to see what normal RGB cameras miss.

- **Hyperspectral Sensing:** The unit must utilize the AS7265x sensor cluster (a triad of optical sensors) to read 18 distinct spectral bands across UV, Visible, and NIR (350-1000 nm). This allows for "spectral fingerprinting" of materials. Unlike a camera that just sees "white floating object," this sensor can distinguish between a white plastic bag (polymer absorption) and white foam (scattering).
- **Chlorophyll Indexing:** The system must calculate the NIR/Red reflectance ratio in real-time. Healthy chlorophyll fluoresces in the Near-Infrared spectrum; detecting this peak allows the system to estimate algae density even before it is visible to the naked eye as green scum, serving as an early warning system for eutrophication.
- **Plastic Identification:** The system must differentiate between water, organic matter (wood/leaves), and polymers (PE, PP, PET). This is achieved by analyzing specific absorption features in the NIR spectrum that are unique to synthetic polymers (C-H bonds).
- **Visual Validation:** A secondary standard RGB camera must capture context images. If the spectral sensor detects an anomaly, the camera snaps a photo to allow human operators to verify the object visually (e.g., distinguishing between a plastic bottle and a floating bird).

6.2 Autonomous Navigation & Mobility

- **GPS Waypoints:** The robot must accept a list of coordinates (Lat/Long) and navigate to them sequentially using the NEO-6M GPS module, maintaining a positional accuracy of within 2.5 meters. It must handle path corrections to account for wind drift and currents.
- **Propulsion:** Twin BLDC motors arranged in a differential steering configuration (tank steering). This eliminates the need for a rudder, which is prone to jamming in aquatic weeds, and allows for a zero-turning radius, essential for maneuvering in tight, stagnant water bodies.
- **Stability:** A catamaran or pontoon-style hull design (constructed from PVC pipes or 3D Printed ABS) is required. This wide-stance design provides superior stability against rolling from ripples and waves, ensuring the optical sensors remain perpendicular to the water surface for accurate readings.
- **Turbine Integration:** A Savonius vertical-axis wind/water turbine should be integrated. Its primary function is to harvest ambient kinetic energy to assist with battery life or, alternatively, to mechanically drive sensor wipers to keep the optical window clean of biofilm.

6.3 Data Processing & Connectivity

- **Edge Computing:** Preliminary classification (Plastic vs. Algae vs. Water) must happen on-board the device (Arduino Mega/Pi Zero). Transmitting raw spectral data is bandwidth-intensive and drains battery; the bot should process the data locally using lightweight ML models (SVM or Logistic Regression) and only transmit the *result*.
- **Telemetry:** Data packets must be transmitted via MQTT protocol over LoRa (for long range rural use) or Wi-Fi (for urban testing). MQTT is chosen for its lightweight overhead, ensuring message delivery even in areas with spotty network connectivity.
- **Dashboard:** A web-based interface (React/Streamlit) displaying:
 - **Live Heatmaps:** Visual overlays of pollution intensity on a map, showing color-coded gradients (Green=Safe, Red=Toxic).
 - **Real-time Status:** Telemetry data including battery voltage, current location, and connection strength.

- **Alerts:** Automated notifications (SMS/Email) for "Hotspots" where pollution exceeds safety thresholds.

7. Technical Architecture

7.1 Hardware Stack

Component	Specification	Purpose
Microcontroller	ESP32 / Arduino Mega / Pi Zero W	Acts as the central brain for control logic, sensor fusion, and connectivity.
Spectral Sensor	AS7265x (18-channel)	The primary data gathering tool; provides the 18-channel spectral fingerprint.
GPS	NEO-6M	Handles location tracking, geofencing, and autonomous path planning.
Motors	2x BLDC Waterproof	Provides differential thrust for propulsion and steering without moving rudders.
Power	5000mAh Li-ion + Savonius Turbine	Main power source with a regenerative assist system to extend runtime.
Chassis	ABS/PETG 3D Printed + PVC Floats	Provides IP67 waterproofing for electronics and necessary buoyancy/stability.

7.2 Software Stack

- **Firmware:** Written in Arduino C++ or MicroPython. Handles motor mixing algorithms (PID control) and sensor interrupt management.
- **ML Engine:** Python (Scikit-learn) running on the edge or cloud. The models are trained on a database of spectral signatures for water, algae, and common plastics.
- **Cloud/Backend:** AWS IoT Core or Google Firebase for real-time database storage and user authentication.
- **Frontend:** Streamlit or React.js with Folium/Leaflet integration for interactive map-based data visualization.

8. Non-Functional Requirements (NFRs)

- **Durability:** The hull and electronics enclosure must be ruggedized to withstand aquatic environments. This includes IP67 waterproofing for the central processing unit and corrosion-resistant coatings for exposed metal parts to survive humidity and splashes.
- **Battery Life:** The system must achieve a minimum continuous operation time of 2 hours on a single charge under standard load conditions. This allows for significant area coverage (approx. 2-3 km track length) per deployment cycle.
- **Latency:** The data update rate from the robot to the dashboard should be < 5 seconds, ensuring that the "Real-Time" claim is valid for operational decision-making during emergency response scenarios.
- **Maintainability:** The design must be modular. Sensors should be connected via standard headers (I2C/UART) to allow for easy swapping or upgrading (e.g., adding a pH sensor) without needing to rebuild the entire chassis.

9. Risks & Mitigation Strategies

Risk	Probability	Mitigation Strategy
Sensor Noise (Sun Glint)	High	Sunlight reflecting off waves can distort spectral readings. Mitigation: Implement spectral averaging (taking multiple readings per second) and use polarizing filters or shrouds over the sensor window to block oblique light.
Water Leakage/Damage	Medium	Water ingress is a constant threat to electronics. Mitigation: Double-seal the electronics box with silicone gaskets and use magnetic reed switches for power to avoid drilling physical holes for buttons.
Debris Entanglement	Medium	Aquatic weeds and fishing lines can jam motors. Mitigation: Design and 3D print specific motor guards/meshes to shield the propellers while allowing water flow, similar to a cage fan.

10. Future Scope (Roadmap)

- **Phase 2 (Chemical Sensing):** Integration of electrochemical sensors for pH, Turbidity, and Dissolved Oxygen to create a comprehensive Water Quality Index (WQI).
- **Phase 3 (Extended Autonomy):** Integration of a solar panel roof to trickle charge the battery, potentially allowing for dawn-to-dusk continuous operation without swapping batteries.
- **Phase 4 (Swarm Logic):** Developing inter-robot communication protocols (Mesh Networking) so multiple ECO-FLOATs can coordinate to map a large lake collaboratively without redundant overlap or collisions.

11. Success Metrics

- **Accuracy:** Achieve >85% correlation between the robot's spectral sensor readings and traditional lab-tested samples for Chlorophyll-a concentration.